

LAB ASSIGNMENT – 4.2

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BATCH – 29

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TASK-1: Zero-shot: Prompt AI with only the instruction. Write a Python function to determine whether a given number is prime

CODE:

```
# Zero-shot Prompt AI with only the instruction Write a Python function to determine whether a given number is prime
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True
# Example usage
print(is_prime(11))
```

OUTPUT: TRUE

```
PS C:\Users\Shree priya\Downloads> & 'c:\Users\Shree priya\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\Shree priya\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '50174' '--' 'c:\Users\Shree priya\Downloads\aac4.2.py'
True
```

TASK-2: One-shot: Provide one example: Input: [1, 2, 3, 4], Output: 10 to help AI generate a function that calculates the sum of elements in a list.

CODE:

```
# One-shot: Provide one example: Input: [1, 2, 3, 4], Output: 10 to help AI generate a function that calculates the sum of elements
def sum_of_list(lst):
    return sum(lst)
print(sum_of_list([1, 2, 3, 4]))
```

OUTPUT : 10

TASK-3: Few-shot: Give 2–3 examples to create a function that extracts digits from an alphanumeric string.

CODE:

```
# Few-shot: Give 2-3 examples to create a function that extracts digits from an alphanumeric string
def extract_digits(s):
    return ''.join(filter(str.isdigit, s))
print(extract_digits("a1b2c3"))
```

OUTPUT: 123

TASK-4: Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string.

CODE:

```
# Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string
def count_vowels(s):
    vowels = 'aeiouAETOU'
    return sum(1 for char in s if char in vowels)
print(count_vowels("Hello World"))
```

OUTPUT: 3

TASK-5: Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers without using the built-in min() function.

CODE:

```
# Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers
def min_of_three(a, b, c):
    if a <= b and a <= c:
        return a
    elif b <= a and b <= c:
        return b
    else:
        return c
print(min_of_three(3, 1, 2))
```

OUTPUT:1